

#Atmospheric Dynamics



Chapter one: Introduction



Objectives

- To define the Atmospheric dynamics & Atmospheric Continuum
- To Be familiar with Physical Dimensions, Units and Dimensional Homogeneity
- To Get the Idea Of Dimensional (Scale) Analysis

Chapter one: Introduction

The atmospheric Continuum

- Atmospheric dynamics is the study of those motions of the atmosphere that are associated with weather and climate.
- The atmosphere can be regarded as a continuous fluid medium, or *continuum*.
- *Air parcel* or *air particle* are terms used to refer to a point in the continuum, which is a volume element that is very small compared with the volume of atmosphere under consideration.
- Air parcel still contains a large number of molecules.

The atmospheric Continuum

- The various physical quantities that characterize the state of the atmosphere (e.g., pressure, density, temperature) are assumed to have unique values at each point in the atmospheric continuum.
- The field variables (P , ρ , T) and their derivatives are assumed to be continuous functions of space and time.

The atmospheric Continuum

- The fundamental laws governing the motions of the atmosphere are expressed in terms of partial differential equations involving the field variables as dependent variables and space and time as independent variables.
- The motions of the atmosphere are governed by the fundamental physical laws of conservation of *mass*, *momentum*, and *energy*. These laws are applied to air parcel to obtain the governing equations.

Physical Dimensions and Units

❑ A **dimension** is a measure of a physical quantity without numerical values, while a **unit** is a way to assign a number to the dimension.

❑ For example, length is a dimension, but centimeter is a unit.

Primary dimensions and their associated primary SI and English units

Dimension	Symbol*	SI Unit	English Unit
Mass	m	kg (kilogram)	lbm (pound-mass)
Length	L	m (meter)	ft (foot)
Time [†]	t	s (second)	s (second)
Temperature	T	K (kelvin)	R (rankine)
Electric current	I	A (ampere)	A (ampere)
Amount of light	C	cd (candela)	cd (candela)
Amount of matter	N	mol (mole)	mol (mole)

Exercise

☐ **The Primary Dimensions of Force?**

DIMENSIONAL HOMOGENEITY

- ❑ We've all heard the old saying, You can't add apples and oranges
this is actually a simplified expression of a far more global and
fundamental mathematical law for equations,
- ❑ The **law of dimensional homogeneity**, stated as
- ❑ “Every additive term in an equation must have the same
dimensions.”

DIMENSIONAL HOMOGENEITY

- ❑ The fundamental laws that govern the motions of the atmosphere satisfy the principle of *dimensional homogeneity*.
- ❑ That is, all terms in the equations expressing these laws must have the same physical dimensions.
- ❑ These dimensions can be expressed in terms of multiples and ratios of four dimensionally independent properties: **length, time, mass, and thermodynamic temperature.**
- ❑ To measure and compare the scales of terms in the laws of motion, a set of units of measure must be defined for these four fundamental properties.

Exercise

- ❑ **Derive Dimensional Homogeneous equation for the change in total energy**

Change of total energy of a system:

$$\Delta E = \Delta U + \Delta KE + \Delta PE$$

SCALE ANALYSIS

- The process of simplifying the equations of motion using typical scales for the phenomenon of interest is called *scale analysis*.
- Scale analysis, or scaling, is a convenient technique for estimating the magnitudes of various terms in the governing equations for a particular type of motion.
- The general set of partial differential equations governing the motions of the atmosphere is extremely complex; no general solutions are known to exist.

SCALE ANALYSIS

- To acquire an understanding of the physical role of atmospheric motions in determining the observed weather and climate, it is necessary to develop models based on systematic simplification of the fundamental governing equations.
- particular atmospheric motion systems requires careful consideration of the scales of motion involved

SCALE ANALYSIS

- ❖ The atmosphere is a complex system capable of supporting many different types of motion from small turbulent eddies between two buildings to large weather systems.
- ❖ In order to make the theoretical study of a particular system easier, we can simplify our approach by knowing which scales are important in the driving mechanisms of that system.
- ❖ This allows us to determine whether we can ignore processes that are happening at different scales.
- ❖ Time scale is important as is length scale.

SCALE ANALYSIS

How do we determine these scales?

- ✓ For a particular type of system, we measure the magnitudes of the important variables, such as pressure, wind speed, and moisture mixing ratio.
- ✓ From these we determine the amplitudes and typical distances over which fluctuations occur.
- ✓ If we are looking at a wave-like phenomenon, we can measure variables like frequency and wavelength.

SCALE ANALYSIS

- ❑ From a series of measurements of many different examples of the same type of system, we can determine typical values for many of these quantities, and hence determine the appropriate scale.
- ❑ Some scales of some common phenomena are shown in Table
- ❑ These typical values are then used to compare the magnitudes of various terms in the governing equations.

Type of motion	Horizontal length scale	Time scale
Cold front	50 km	1–2 days
Tornado	100 m	Minutes
Mid-latitude weather system	1000 km	Several days
Cumulus cloud	1 km	Tens of minutes
Surf	10 m	Seconds
Planetary wave	10 000 km	Weeks–months
European Alps foehn or Rocky Mountain chinook	10 km	Hours

Scale	Symbol	Magnitude
Horizontal wind scale	U	10 m s^{-1}
Vertical wind scale	W	10^{-2} m s^{-1}
Horizontal length scale	L	10^6 m
Vertical length scale (depth of troposphere)	H	10^4 m
Time scale (L/U)	T	10^5 s
Kinematic viscosity	ν	$10^{-5} \text{ m}^2 \text{ s}^{-1}$
Dynamic pressure scale	$\delta p / \rho$	$10^3 \text{ m}^2 \text{ s}^{-2}$
Total pressure scale	P / ρ	$10^5 \text{ m}^2 \text{ s}^{-2}$
Gravity	g	10 m s^{-2}
Density variation scale	$\delta \rho / \rho$	10^{-2}

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Thanks !

#Atmospheric Dynamics

Chapter one: Introduction



Objectives

- To Be aware with Meteorological conventions that used to define wind in the atmosphere.
- To know the coordinate system and To Be familiar with Types of coordinate system
- To See the Conversion from Cartesian to Spherical coordinate system.
- To Get the Idea Of Types of control volume which are commonly used to examine the motion of fluid elements.

Meteorological Conventions That Used To Define Wind In The Atmosphere

Zonal

Direction oriented along lines of latitude (west-east or east-west component)

East: Positive

West: Negative

Wind Component: u



Westerly Wind

$$u > 0$$



Easterly Wind

$$u < 0$$

Meridional

Direction oriented along lines of longitude (south-north or north-south component)

North: Positive

South: Negative

Wind Component: v



Southerly Wind

$$v > 0$$



Northerly Wind

$$v < 0$$

Vertical

Direction oriented along the vertical axis

Upward: Positive

Downward: Negative

Wind Component: w



Upward Wind

$$w > 0$$



Downward Wind

$$w < 0$$

(Top-Down Perspective)

$$\vec{v} = u\hat{i} + v\hat{j} + w\hat{k}$$

Coordinate systems

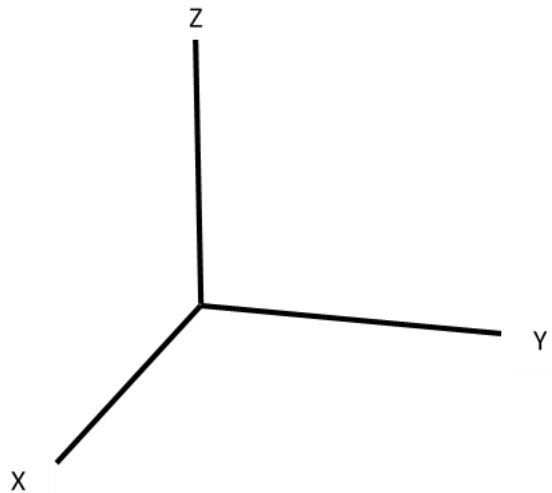
- ❑ A coordinate system is an artificial mathematical tool that we construct in order to describe the position of a real object.
- ❑ Coordinate systems are used to describe the position of an object in space.
- ❑ Longitude and latitude are a special kind of coordinate system, called a **spherical coordinate system**, since they identify points on a sphere or globe.

Coordinate systems

- ❑ Most coordinate systems use two numbers, a **coordinate**, to identify the location of a point.
- ❑ Each of these numbers indicates the distance between the point and some fixed reference point, called the **origin**.
- ❑ The first number, known as the X value, indicates how far left or right the point is from the origin.
- ❑ The second number, known as the Y value, indicates how far above or below the point is from the origin.
- ❑ The origin has a coordinate of 0, 0.

Horizontal Coordinate Systems

- The most widely used coordinate system in atmospheric modeling equations are *Cartesian (rectangular)* and *spherical* coordinate systems
- Cartesian (rectangular) coordinates are used on the microscale and mesoscale to simulate flow.



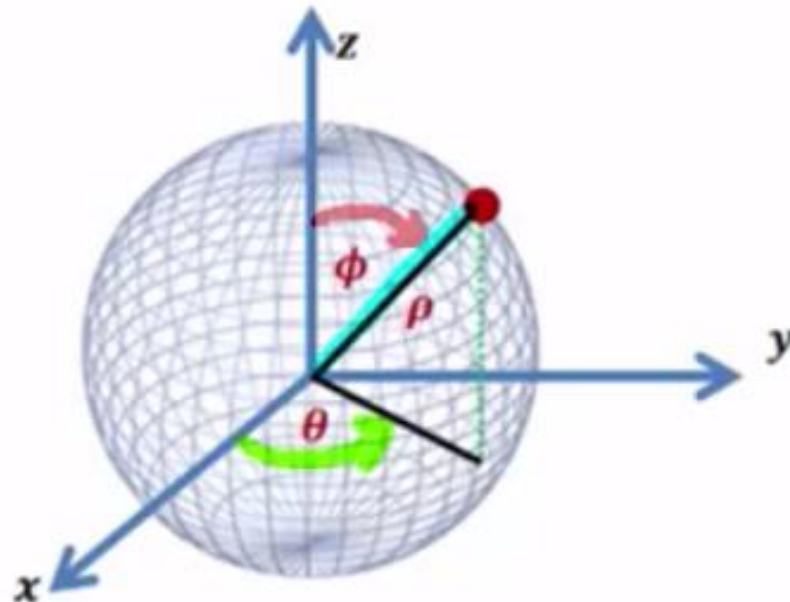
Cartesian Coordinate system

- Over long distances, curvature of the earth prevents the accurate division of the earth into contiguous set of rectangles

Spherical Coordinates

In a spherical coordinate system, a point P in space is represented by an ordered triple (ρ, θ, ϕ) .

- ρ is the distance between the point and the origin. ($\rho \geq 0$)
- θ is angle counter clockwise from the pole or positive x-axis in the xy-plane. ($r \geq 0$)
- ϕ is the angle between the positive z-axis and the point. ($0 \leq \phi \leq \pi$)



Spherical coordinate system

0^0 Longitude...Greenwich

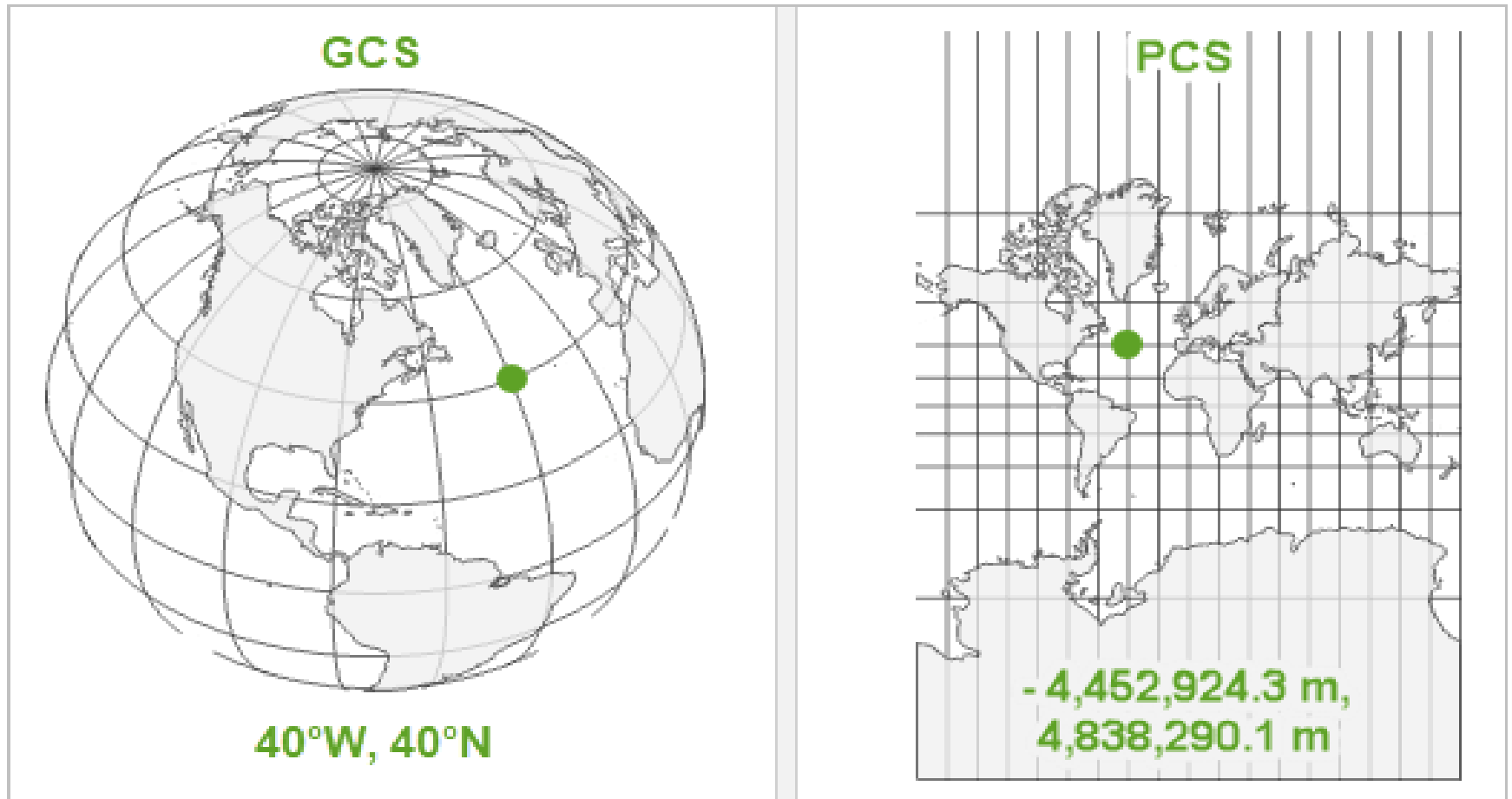
0^0 Latitude....equator

- The spherical coordinate system divides the Earth into **longitudes** (meridians), which are S-N lines, and **latitudes** (parallels), which are W-E lines parallel to each other.
- Advantage: it takes curvature of the earth in to account.

Geographic Vs. projected coordinate system

- A geographic coordinate system (GCS) is a reference framework that defines the locations of features on a model of the earth. It's shaped like a globe—spherical. Its units are angular, usually degrees.
- A projected coordinate system (PCS) is flat. defines the locations of features on a flat surface. Its units are linear, most commonly in meters.

Geographic Vs. projected coordinate system



Exercise

- ❑ *Conversion from Cartesian to Spherical ?*
given (2,3,5)

Ans (6.16, 56.3, 35.73)

Reference Frames

- There are two types of reference frames:
 1. Inertial Reference Frame
 2. Non-Inertial Reference Frames
- *Geocentric* reference frame is a frame of reference at rest with respect to the rotating earth.
- Newton's first law of motion states that a mass in uniform motion relative to a coordinate system fixed in space will remain in uniform motion in the absence of any forces. Such motion is referred to as *inertial motion*; and the fixed reference frame is an inertial, or absolute, frame of reference.
- An object at rest or in uniform motion with respect to the rotating earth is not at rest or in uniform motion relative to a coordinate system fixed in space. Therefore, motion that appears to be inertial motion to an observer in a geocentric reference frame is really accelerated motion.

Inertial Reference Frame

- ❑ Inertia is the property of a body by the virtue of which it either remains at rest or in the state of motion until it is acted upon by any external force.
- ❑ Inertial frame of reference takes into account this property.
- ❑ Every body which is analyzed under an inertial frame of reference tends to maintain its position if rest or of motion until the body is acted upon by any external force.
- ❑ Thus, it can be said that an inertial frame of reference either remains at rest or moves with a constant velocity.
- ❑ As long as the frame of reference is not moving or moving with a constant velocity it is termed as an inertial frame of reference.

Non-Inertial Reference Frames

- ❑ A non-inertial frame of reference is one which is in the state of acceleration.
- ❑ If a body seems to be not obeying the inertia rules, it can be said to be in a non-inertial frame of reference. In this frame of reference, Newton's first law of motion does not hold true.
- ❑ For example, if a car is moving with a certain acceleration it is said to be in a non-inertial frame of reference.
- ❑ . If the frame is accelerating or moving in a circular path with constant speed, it is termed as a non-inertial frame of reference

Rotating Reference Frame

- ❑ The earth rotates with an angular velocity Ω which we will take, for our purposes to be constant with time although including a time variation for Ω is not difficult and may be necessary for dynamics on very long geological time scales. The Earth's rotation rate is $7.29 \times 10^{-5} \text{ sec}^{-1}$ (once per day).
- ❑ The radius of the Earth, R , is about 6,400 km. This gives an equatorial speed of about 467 m/sec or 1050 miles/hr.
- ❑ There is no current or wind system on the Earth that approaches speeds of that magnitude. To the lowest order the ocean and atmosphere are moving with the rotation rate of the planet and all the currents and winds that we observe are very small departures from the solid body rotation of the Earth.

Rotating Reference Frame

- ❑ It is therefore sensible that we want to describe the winds and currents that we see from within a coordinate frame that removes the basic rotation and shares with us the observational platform of the rotating Earth.
- ❑ A rotating reference frame is a coordinate system that is in motion due to rotation relative to an inertial (non-accelerating) reference frame.

- In meteorology, a rotating reference frame is often used to describe and analyze atmospheric motion, especially on a large scale such as weather systems and global circulation patterns.
- The Earth's rotation plays a significant role in shaping atmospheric phenomena, and understanding these dynamics requires a rotating reference frame.

Vertical Coordinate System

- Altitude (geometrical height) is the common vertical coordinate for both cartesian and spherical coordinates
- Pressure Coordinate System
- Sigma Coordinate System

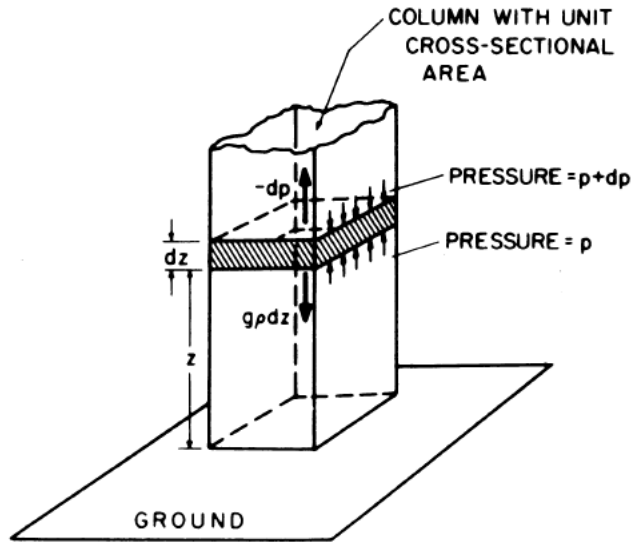
Pressure as a vertical coordinate system

- The thermodynamic state of the atmosphere at any point is determined by the values of pressure, temperature, and density (or specific volume) at that point.
- These field variables are related to each other by the equation of state for an ideal gas.
- Letting p , T , ρ , and $\alpha(\equiv 1/\rho)$ denote pressure, temperature, density, and specific volume, respectively, we can express the equation of state for dry air as

$$p\alpha = RT \quad \text{or} \quad p = \rho RT \dots\dots\dots \text{ideal gas law}$$

where R is the gas constant for dry air ($R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$).

Pressure as a vertical coordinate system



The Hydrostatic Equation

$$\frac{dp}{dz} = -\rho g$$

Integrating hydrostatic equation from a height z to the top of the atmosphere:

$$p(z) = \int_z^{\infty} \rho g dz$$

- It is often useful to express the hydrostatic equation in terms of the geopotential rather than the geometric height.

$$dP = -\rho g dz$$

$$\alpha = RT/p$$

$$\text{Density} = P/RT$$

$$g dz = -\alpha dP = -\left(\frac{RT}{P}\right) dP = -RT d \ln P \quad \dots\dots\dots (dP/P = d \ln P)$$

Pressure as a vertical coordinate system

Thus, the variation of geopotential with respect to pressure depends only on temperature

$$\Phi(z_2) - \Phi(z_1) = g_0(Z_2 - Z_1) = R \int_{p_2}^{p_1} T d \ln p$$

- Here $Z = (z)/g_0$, is the *geopotential height*, where $g_0 \equiv 9.80665\text{ms}^{-2}$ is the global average of gravity at mean sea level.

$$Z_T \equiv Z_2 - Z_1 = \frac{R}{g_0} \int_{p_2}^{p_1} T d \ln p$$

where Z_T is the *thickness* of the atmospheric layer between the pressure surfaces p_2 and p_1

Sigma as a vertical coordinate

- It is usually assumed that the height of the ground z_0 is coincident with the pressure surface p_s (where p_s is usually set equal to 1000 hPa).
- These assumptions are not strictly valid. Pressure changes at the ground, the height of the ground generally varies (variation in topography) and hence the lower boundary condition should not be applied at a constant p_s .
- The *sigma coordinate* is defined as $\sigma = p/p_s$, where $p_s(x, y, t)$ is the pressure at the surface.
- Thus, σ is a nondimensional vertical coordinate that decreases upward from a value of $\sigma = 1$ at the ground to $\sigma = 0$ at the top of the atmosphere.

How to examine the motion of fluid element?

- ❑ There are two point of view from which we can examine the motion of fluid element.

1. *Lagrangian*

2. *Eulerian*

- ❑ A fluid element is very small parcel of fluid. Although there is no official size of a fluid element think of it as a chunk of fluid that is smaller than your eyes can see, but is still made up of many molecules.

The *Lagrangian* Frame

- ❑ In the *Lagrangian* frame, the control volume moves about following the motion of the fluid, always containing the same fluid particles.
- ❑ The individual fluid element are traced in time.
- ❑ Uses a position vector that changes in time. The parcel position is a function of its starting point, so the history of a parcel is known.

The *Lagrangian* Frame

- Benefits:
 - Useful for developing theory
 - Very powerful for visualizing fluid motion
 - The history of each fluid parcel is known
- Problems:
 - Need to keep track of a coordinate system for each parcel
 - How do you account for interactions between different parcels?
 - What can we say about regions where fluid parcels are sparse?
 - What can we say about the fluid as a whole if all of the initial parcels bunch together?

The *Lagrangian* Frame

The Lagrangian Frame

Weather balloons are a source of measurements in the Lagrangian frame. They are passively transported by the background wind field (and so are essentially isolated fluid parcels).

Eulerian frame of reference

- ❖ In the *Eulerian* frame of reference, the control volume consists of a parallelepiped of sides δx , δy , δz , whose position is fixed relative to the coordinate axes.
- ❖ Fluid properties are observed at point in space.
- ❖ Measurement could be taken at a **single point** as a function of time. The global atmosphere is quantified by using an amalgamation of this individual measurement.
- ❖ In *Eulerian* frame of reference the properties of the whole atmosphere are described in terms of individual measurement taken at affixed location over time.

The Eulerian Frame

Under the ***Eulerian*** point of view.

- **Benefits:**
 - Useful for developing theory
 - Requires considering only one coordinate system for all parcels
 - Easy to represent interactions of parcels through surface forces
 - Looks at the fluid as a field.
 - A value for each point in the field – no gaps or bundles of “information.”
- **Problems**
 - More difficult to keep track of parcel history—not as useful for applications such as pollutant dispersion (or clouds?)...

The Eulerian Frame



Weather stations are a source of measurements in the Eulerian frame. They are placed in a fixed location and observe the properties of fluid parcels as they pass by the instruments.

Question: Why consider two frames of reference?

Answer: Since the same physical principles hold regardless of the reference frame, the use of multiple reference frames is primarily for purposes of understanding.

Certain concepts can be more easily explained in the Lagrangian frame, whereas others are better explained in the Eulerian frame.

1. What is atmospheric Dynamics?

- A) The study of the physical properties of the atmosphere
- B) The study of the chemical reactions in the atmosphere
- C) The study of the motion of the atmosphere
- D) The study of the interactions between the atmosphere and the Earth's surface

2. Which one of the following is wrong

- A. Temperature Is Dimension And Kelvin Is Unit
- B. Unit Is A Way To Assign A Number To The Dimension
- C. A Dimension Is A Measure Of A Physical Quantity With Numerical Values
- D. The Fundamental Laws That Govern The Motions Of The Atmosphere Satisfy the Principle Of Dimensional Homogeneity

3. The process of simplifying the equations of motion using typical scales for the phenomenon of interest is called

A. Perturbation method

B. Scale analysis

C. Dimensional homogeneity

D. Lagrangian method

4. From the following one is not a Meteorological convention that used to define wind in the atmosphere

A. Zonal

B. Vertical

C. Meridional

D. Eulerian

5. What is the Spherical equivalent of a Cartesian **(2,3,5)**

A. 5.33, 45.3,30.7

B. 6.16, 55.39,38.7

C. 3.56, 57.37,55.83

D. 6.16, 56.3,35.7

6. From the following one is **NOT** a fundamental force

A. Pressure gradient force

B. Coriolis force

C. The gravitational force

D. The friction force

7. Which one of the following is an example of the Eulerian frame of reference

A. Meteorological Stations

B. Aircraft

C. Meteorological balloon

D. All

8. What are the fundamental physical principles that govern Atmospheric motions
- A. conservation of soil, conservation of momentum, and conservation of energy
 - B. conservation of mass, conservation of momentum, and conservation of energy
 - C. conservation of water, conservation of momentum, and conservation of energy
 - D. all

....END

Thanks !